

Cambridge International AS Level
Information Technology 9626 · Paper 1 Theory

Topic 10

Database and file concepts

35 questions from 23 question papers, Feb/March 2022 to May/June 2026.

Questions come first, in date order, then their mark schemes in the same order. Each question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, or use the bookmarks panel.

A question that covers more than one topic appears in each of those topics. Topics follow the 2025-2027 syllabus.

Questions

9626/12 Feb/March 2022 · Question 10

[Mark scheme](#)

10 (a) Define the following database terms.

(i) One-to-one relationship

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..... [1]

(ii) One-to-many relationship

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..... [1]

(iii) Compound key

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..... [1]

(b) Explain how a many-to-many relationship can be created in a database.

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..... [3]

[8]

[6]

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1 (a) Describe the features and the role of a primary key in a database.

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(b) Describe the features and the role of a foreign key in a database.

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5 Compare static parameter queries with dynamic parameter queries. You must include similarities and differences.

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- 10** A school stores all the records of its students in alphabetical order. Records could be found using sequential access.

(a) Explain what is meant by sequential access.

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..... [1]

The school decides to use the indexed sequential method to search for the records of individual students. An index table contains all the letters of the alphabet in alphabetical order.

(b) Describe how the details of a student called Johnsen would be found.

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When she created the database, she considered creating a dynamic parameter query to produce her printout. However, she decided to create a static parameter query.

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[4]

9626/13 Oct/Nov 2023 · Question 6

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2 A data dictionary is required when creating a relational database.

Identify and describe **four** components that should be included in a data dictionary.

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11 Explain what is meant by the term open-source file format.

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[8]

- 1** A building company has a relational database. The database has two tables: CUSTOMERS and QUOTATIONS.

- (a)** The QUOTATIONS table has a foreign key.

Describe what is meant by the term foreign key.

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- (b) (i)** Define the term referential integrity.

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- (ii)** Explain why referential integrity is important.

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(c) Describe what is meant by the term relational database.

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9626/13 May/June 2024 · Question 1

Mark scheme

1 A relational database has been created for customer orders.

The entities and attributes in the relational database are:

CUSTOMER	ORDER	PRODUCT
Cust_ID	Order_code	Product_ID
First_name	Cust_ID	Description
Family_name	Date	Price
Address_1	Product_ID	
Town	Quantity	
Zip_code		

(a) Identify the primary key in the CUSTOMER entity.

..... [1]

(b) Identify the primary key in the ORDER entity.

..... [1]

(c) Identify **one** foreign key in the ORDER entity.

..... [1]

(d) Identify the relationship between the CUSTOMER and ORDER entities.

..... [1]

[8]

[6]

Mark scheme

(a) Describe the following database relationships:

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(b) Describe how a many-to-many relationship could be created in a database.

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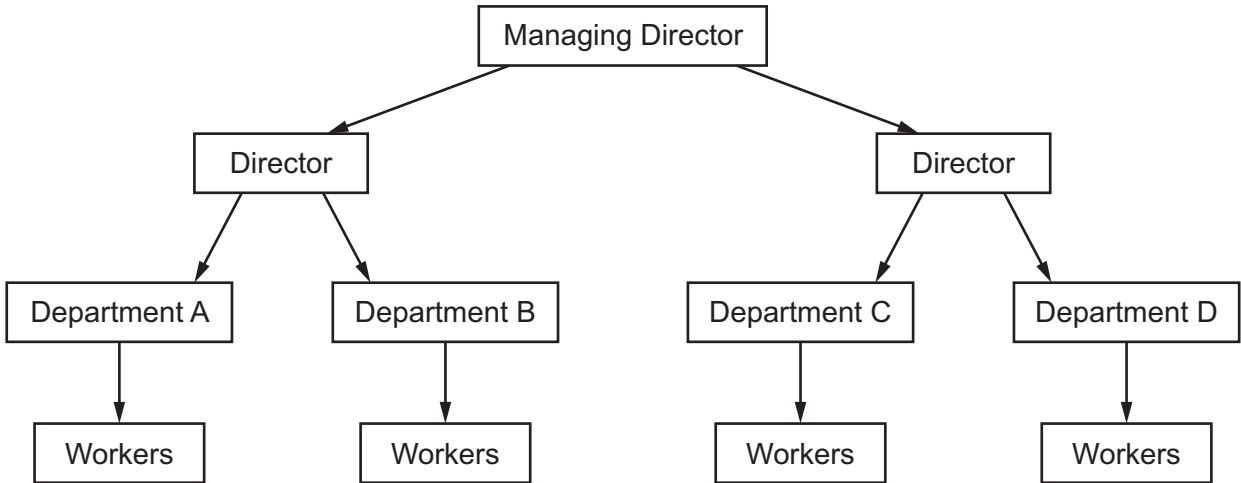
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1 A company uses the following hierarchical staffing structure.



The company stores all its staff records in a hierarchical database.

(a) Explain what is meant by a hierarchical database management system.

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(b) Describe how the company’s hierarchical database would be used to store all the staff records.

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8 A furniture company uses databases in its warehouse to help with stock control.

(a) The company has a Product database which contains the fields shown in the table.

Field Name	Data Type	Example of Data
Number_in_Stock		199 (Number_in_Stock can vary between a minimum of 0 and a maximum of 250.)
Location		07.44.03
Reorder_Level		250
Price		\$203.50
Supplier_Telephone_No.		07700900819

Complete the Data Type column in the table by writing the most appropriate data type for each field. For any numeric fields, specify the numeric data type. [5]

- (b)** State and describe appropriate validation checks that could be carried out on data as it is entered into the Product database. You can describe different validation checks for the same field.

[6]

- (c) As well as the Product database, the company has a Customer database. Describe how the two databases can be combined into one relational database.

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3 Normalisation is used in a relational database.

(a) State what is meant by the term second normal form (2NF).

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(b) Discuss the benefits and drawbacks of normalising data.

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10 Describe the following types of database query:

(a) summary

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(b) complex

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(c) nested

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[Mark scheme](#)

- 9 Eric has made a spreadsheet that he has shared with a number of friends. Some of his friends cannot open the spreadsheet.

Identify **three** possible reasons the spreadsheet will not open once received. For each reason, suggest a way Eric could have prevented the problem.

Reason 1

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Way 1

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Reason 2

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Way 2

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Reason 3

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Way 3

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[6]

9626/13 May/June 2025 · Question 12

[Mark scheme](#)

- 12 Define the following types of database key fields:

(a) Composite key

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..... [1]

(b) Compound key

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..... [1]

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[Mark scheme](#)

7 (a) Describe the term dynamic parameter query.

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(b) Describe the benefits of using dynamic parameter queries when using a database.

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[Mark scheme](#)

15 Discuss the importance of referential integrity.

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7 A company wishes to gather some information from employees.

(a) Describe the process of organising face-to-face interviews.

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(b) Justify the use of a relational database to store the information gathered.

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- 10** A list of some field names from a database is shown. For each field name, identify the most appropriate data type.

(a)

Field name	Most appropriate data type
employee_name	
hourly_rate_of_pay	
was_employee_born_in_france?	

[3]

- (b) A presence check has already been run on each field of the database.

For each of the following fields, identify and describe **one** other type of validation that can be used. You should only use each type of validation once.

- (i) hourly_rate_of_pay

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- (ii) date_of_birth

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- (c) Justify why a field holding the age of each employee has been set as a calculated field.

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- 5 A school has paper-based records of all students that have attended the school since 1993.

These records include:

- home addresses and contact details
- the grade each student achieved in each of their end of year exams
- the years when they joined and left the school.

The school is going to create and use a single database to store these records.

- (a) Compare the use of flat file database structures and relational database structures to store these records.

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The database will include validation rules.

- (b) Identify and describe **one** validation rule that could be applied to the database. State why it is important.

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9 A student is planning a relational database.

(a) Define the term relational database.

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The relational database could include both primary keys and compound keys.

(b) (i) Describe the characteristics of a primary key in a relational database.

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(ii) Describe the role of a primary key in a relational database.

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..... [1]

(c) Describe how a compound key differs from a primary key.

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The database the student is planning needs to be normalised.

(d) Describe the conditions that must be met for the database to be in first normal form (1NF).

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(e) Describe **four** benefits of normalising the relational database.

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Mark schemes

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Question	Answer	Marks
10(a)(i)	Both tables can have only one record on either side of the relationship	1
10(a)(ii)	A table contains individual records that each relate to many records in the related table	1
10(a)(iii)	A compound key is a primary key that combines more than one foreign key to make a unique value	1
10(b)	Three from: A many-to-many relationship has to be broken down into two/several one-to-many relationships... ...using a third table, a “join table”... ...each record in the “join table” would have the foreign key fields... ...linked to the primary keys of the two tables it is joining together	3

Mark scheme · 9626/13 May/June 2022 · Question 10

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Question	Answer	Marks
10	Eight from: <i>Advantages:</i> Any change to one record which is needed can be instantly made to any related records The database does not have data duplicationmaking the file size smallerso less money needs to be spent on storage ...so there are fewer errors in the data... ...so there is less chance of storing incorrect copies of the data Modifying a table is easier as there is less data to modify <i>Disadvantages:</i> A larger numbers of tables requires more relationships to be designed... ...taking more time Making data atomic may not always be the best solution such as date of birth can be separated into day, month and year... ...but this may serve no purpose Data may be stored as codes rather than meaningful data... ...making it difficult for humans to interpret/level of detail can be lost With more tables, setting up queries can become more difficult... ...the more complex the database queries need to be so can take longer to run You can end up with more tables than an unnormalised database making it difficult to keep track of data May require greater expertise/knowledge and understanding... ...which may need to come from outside and be paid for Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions to be a proper discussion	8

Mark scheme · 9626/13 Oct/Nov 2022 · Question 4

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Question	Answer	Marks
4	Six from: • Data resources/(a) database(s) (Accept any indication of the type of data being held e.g. Employee data/information) (1) • Hardware resources of a system (1) • People management and project management applications (1) • Decision support systems//computer program applications used by managers to gather information to support problem solving//decision making (may exemplify) (1) • Executive information systems/a reporting tool that provides quick access to summarised reports (coming from all departments) (1) • Marketing Information Systems/systems designed specifically for managing the marketing aspects of the business (1) • Accounting information systems for accounting functions (1) • Human resource management systems//systems used for analysing personnel aspects of an organisation (1)	6

Mark scheme · 9626/12 Feb/March 2023 · Question 10

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Question	Answer	Marks
10	Max four from: • A format used for storing data, for anyone to use (1) • An open-source file format can be used by both proprietary and (free and) open-source software (1) • Open source formats are often called free file formats if they are not covered by any copyrights/patents (1) • Anyone may use them at no cost (1) for any desired purpose (1) • Code is open to inspection (1) • Code can be amended without breach of copyright (1) Max three from: • Needed because not everyone can afford proprietary software (1) • Needed so users can work on files in different locations without the need for the same software (1) • Provides a standard file type (1)	5

Mark scheme · 9626/12 May/June 2023 · Question 8

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Question	Answer	Marks
8(a)	Four from: • One mark for identification of any three forms (1) • A method of organising data tables (1) • Attempts to avoid data loss (1) • Attempts to avoid data redundancy//data duplication (1) • Attempts to improve/improves data base efficiency (1) • Breaks data from non-atomic structure to atomic//description of atomisation (takes an example of non-atomic data and states how it becomes atomic) (1) • Breaks data into smaller/other units/tables of data (1) • The units/tables are related to each other (may exemplify) (1) • Only related data is stored in a table (1) • It is an iterative process// It is a multi-step/stage process (1) ... (where) each step/stage improves the efficiency of the database (may be given as a standalone mark) (1)	4
8(b)	Six from: • Searching a large database is quicker to locate records (1) • A relational database does not have duplicated/redundant data (1) • Uses reduced storage space (DNA 'memory') (1) • When data in one field of a table is modified the data in the corresponding field//related field is updated (1) • Integrity constraints ensure that table relationships are/remain valid (1) • It is easier to maintain security of data (1) • Use of related tables reduces need for (as much) data entry (1) • Reduced chance of data entry errors (1) • Easier for users to produce reports (1)	6

Mark scheme · 9626/11 Oct/Nov 2023 · Question 1

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Question	Answer	Marks
1(a)	Three from: - Contains unique data/values (1) - Cannot contain blank data/be empty (1) - Helps to provide quicker searches (1) - It enables relationships/links to be formed between tables (1st) - if a relational database is to be created (1) - There can only be one primary key in a table (1) - Provides a unique identifier for each record (1)	3
1(b)	Two from: - A field that contains data found in a primary key field in/from another table (1) - Links to the primary key of another table (1) - Used to form a relationship between the two tables (1) - Referential integrity is usually implemented through the use of foreign keys (1) - There can be many foreign keys in a table (1)	2

Mark scheme · 9626/11 Oct/Nov 2023 · Question 5

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Question	Answer	Marks
5	Five marks available: Similarities Max 4 - Both search for the data that the user/query requires//that matches the (search) criteria (1) - Both return records containing that data (1) - Both can be used with complex queries (1) Differences Max 4 - A dynamic (parameter) query can be used to search for different values/data each time it is run (1) - In a static query every time that the query is run it will search for the same data/values (1) - Dynamic query needs a value to be entered every time (1) - Static query has the criteria hard coded/does not need to be added each time (1) - Static parameter query would save the time of editing the query every time a different criteria/value was used//Dynamic parameter query would waste the time of editing the query every time a different criteria/value was used (1) - Dynamic parameter query requires more technical knowledge of the user//are more complicated to create (1)	5

Mark scheme · 9626/11 Oct/Nov 2023 · Question 10

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Question	Answer	Marks
10(a)	One from: • Records are accessed in the order they were entered (1) • Each record is read one by one until a match is found (1)	1
10(b)	Three from: • The index table would contain the corresponding part of the disk (1st) • where names beginning with each letter of the alphabet can be found (1) • For names beginning with J the part of the file containing all the As to I's can be ignored/the disk/records is/are accessed where the J's begin (1) • All the records beginning with J will be read one by one until Johnsen is found (1)	3

Mark scheme · 9626/12 Oct/Nov 2023 · Question 7

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Question	Answer	Marks
7	Five from: • She doesn't have to spend time responding to the prompt/typing in the pass mark, (1st) • which Louisa would have to with a dynamic parameter query (1) • because the pass mark is hard coded (1) • The <u>pass</u> mark is always the same/fixed so there is no need for a dynamic parameter query//to type in the (same) number (1) • Louisa would need greater technical knowledge to set up a dynamic parameter query (rather than a static parameter query) (1) • It would have taken Louisa longer to set up a dynamic parameter query (1) • Louisa would have been more likely to have made mistakes setting up a dynamic parameter query (1) • There is an increased chance of entering the wrong number as the pass mark, as she must enter it every time (1)	5

Mark scheme · 9626/12 Oct/Nov 2023 · Question 8

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Question	Answer	Marks
8	Eight from: - Files created in one software package are sometimes difficult to use/open in other software (1st) - particularly if the packages are incompatible (1) - Generic file formats are compatible with different software packages (1) - When transferring a file to a different computer/device it might not load properly (1st) - since generic file formats are needed to enable different operating systems to read the file/because the particular file format will not operate on that OS (1) - Two marks plus extensions for two examples of the use of generic file types (for a total of 4 marks). e.g. - Word processing files should be saved as a .txt file type (1st) - so that database software can understand it (1) - Database files should be saved/exported as a .csv/.txt/.rtf file type (1st) - so that word processing software can understand it (1) Max 6 marks for just bullet points	8

Mark scheme · 9626/13 Oct/Nov 2023 · Question 2

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Question	Answer	Marks
2	Four from: File size is smaller (1) - Takes up less storage (space) (1) - Text format doesn't need to buy a licence/software (1) - As the file size is smaller the processing time is shorter/less processing power is required (1) - Does not need to be opened by a WP package (1) - Can be opened by more applications (1) - Data can be exchanged between different computers/operating systems (1) - Word processing software has different versions and one version may not open documents produced by a different version (1)	4

Mark scheme · 9626/13 Oct/Nov 2023 · Question 6

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Question	Answer	Marks
6	Four from: - Simple query is on one field (1) - Complex query is on more than one field (1) - Complex query can use Boolean operators (1) - One mark for TWO examples of Boolean operators: AND, OR, NOT (1) - Simple query use a single criterion only (1) - Complex query uses more than one criterion (1)	4

Mark scheme · 9626/12 Feb/March 2024 · Question 2

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Question	Answer	Marks
2	Eight available. Max two for each matched pair: Table name/title (1st) ...a unique/meaningful name for each table (in the database)//to give context (1) Field name/title (1st) ...a unique/meaningful name to identify each field//to give context (1) (Candidate may describe a meaningful name by example – e.g. ‘full name to hold first and surname’) Data type (1st) ...Such as (e.g.) integer/alphanumeric Field length/size (1st) ...specify the number of characters (in each field) (1) Field default value (1st) ...the value automatically entered when a new record is created (1) - Description of each field (1st) ...explanation of the content (of each field) (1) - Required/null field (1st) ...whether a field can be left blank (1) - Validation (1st) ...test to be applied to data that is entered (1) ...to ensure data entered is sensible/reasonable (candidates may describe this instead of expressly stating it) (1) - Table security (1st) ...who has access to edit (accept examples, such as write or delete) the table - Primary Keys/Foreign keys (1st) ...to identify unique records (1) ...to link tables (1) - Relationships (1st) ...types of relationship between tables are stated (1)	8

Mark scheme · 9626/12 Feb/March 2024 · Question 11

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Question	Answer	Marks
11	Three from: • It is a format for storing (digital) data • Defined by a publicly published specification • Specification maintained by a standards organisation • Can be used and implemented by anyone • An open-source file format can be used by both proprietary AND free/open-source software • Also called free file formats if they are not covered by any copyrights/patents • They can be used at no monetary cost (for any desired purpose). • Example of open-source format (CSV/TXT/JPG)	3

Mark scheme · 9626/11 May/June 2024 · Question 4

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Question	Answer	Marks
4	Eight from: • Allows records to be accessed in the order they were entered (1) • Allows records to be accessed randomly (1) • Each record is given index (1) • A database may have several indexes (1st) – based on the information required (1) • Each record is given index (1) • It is a method of allowing fast retrieval (1) • Data is organised into records which consist of fixed length fields (1) • Uses a set of hash tables (1st) – contain "pointers" into the records (1) • Individual records can be retrieved without having to search the entire file (1) • Can be searched quickly (1) • Searches use an index (1st) – which will narrow down the records/data to be searched (1) – that section of the file is then searched sequentially to find the record required. (1)	8

Question	Answer	Marks
9	Max six if concentrates on static or dynamic only Positive Max 6 from: - Static parameter query is fixed/doesn't change/always uses the same criteria (1st) - Which is set when the query is (first) written (1) - saves time when running the query (1) - less likely to have an error because no user input required (1) - A dynamic parameter query can be used to search for different values (1st) - each time it is run (1) - which makes it more flexible (search for other cities) (1) - uses a dialogue box for the user to enter the data required (1) - For dynamic query saves time in designing the query as you don't need to create similar queries (1) - Static queries are more user friendly (1st) - Because it's attached to a button/can click on a button to make it run/don't have to enter search criteria (1) Negative Max 6 from: - In a static query every time that the query is run it will search for the same data (1st) - cannot be changed/search for another city (1) - can end up having a lot of different static queries (1) - making it difficult to find the query you want (1) - If different data is to be searched for the user would need to open up the query in design view (1st) - and change the data in the criteria to that required (1) - Dynamic parameter query requires less technical knowledge of the user. (1) - Static queries require increased storage space/file size (allow converse, but award once only) (1) Two from: - Statement of when static parameter queries are the better option (1) - Statement of when dynamic parameter queries are the better option (1)	8

Question	Answer	Marks
1(a)	Two from: - It is a primary key in/from another table (1st) - A field in CUSTOMERS table (1) OR - a primary key that is referenced/used in another table (1st) - A field in CUSTOMERS table (1) - Is used to link/create relationships between tables (1) (may be awarded where candidate has already identified that different tables exist)	2
1(b)(i)	One mark available: - Referential integrity forces table relationships to be consistent (1) Enforces table relationships (1) - Ensures that both sides of the relationship exist (1) - Ensures that any foreign key references an existing primary key (1)	1
1(b)(ii)	Three from: - It avoids/eliminates redundant/unnecessary data (1) - It avoids orphan data (1) (Ensures that) every foreign key value has a matching value in the corresponding primary key// Referential integrity can be used to ensure foreign keys are valid (1) - Referential integrity prevents//warns of the deletion of related records (1) - Value/data within the related fields cannot be changed (1)	3
1(c)	Five from: - A relational database consists of more than one table (1st) - Each table holds data/information about one area/that is related (1) - Tables are linked to each other//relationships are established between tables...(1st) - using primary key (fields) (1) foreign key (fields) (1) compound key (fields) (1) - Using one to one//one to many relationships (1) - Can create queries using more than one table (1) - Can create reports using more than one table (1)	5

Mark scheme · 9626/13 May/June 2024 · Question 1

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Question	Answer	Marks
1(a)	Cust_ID (1)	1
1(b)	Order_code (1)	1
1(c)	One from: - Cust_ID (1) - Product_ID (1)	1
1(d)	One to many//one CUSTOMER to many ORDER(s) (1)	1

Mark scheme · 9626/13 May/June 2024 · Question 10

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Question	Answer	Marks
10	Eight from: - CSV means comma separated value (1) Benefits MAX SIX: e.g. - csv is human readable (1) - can be edited in a text editor (1) - csv is processed by multiple//almost all applications (1) - csv is small in file size//takes up little storage space (1) Drawbacks MAX SIX: e.g. - csv includes data//values only (1) - Does not include formatting (1) - There is no distinction between text and numeric values (1) - No standard way to represent binary data (1) - Problems with importing csv into SQL (no distinction between NULL and quotes) (1) - Poor support of special characters (1) - No standard way to represent control characters (1) - Lack of universal standard. (1)	8

Mark scheme · 9626/11 Oct/Nov 2024 · Question 3

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Question	Answer	Marks
3	SIX from: • MIS is used in decision making (1) • MIS is used to coordinate/control data in a business. (1) • MIS software can provide past, present and predictive (TWO FROM) information (1) • MIS software can be used with project management (1) • MIS system may be based on a database (1st) • which can be queried (1) • Can be used to gather/analyse/assess information (1st) • e.g. personnel/sales/inventory/production/employee performance (1) • MIS software could be used to create reports. (1) • By comparing reports to previous reports, trends can be identified (1) • Can produce graphs/charts (1)	6

Mark scheme · 9626/11 Oct/Nov 2024 · Question 4

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Question	Answer	Marks
4(a)(i)	ONE mark: • Each record in one table relates to only one record in another table (1).	1
4(a)(ii)	ONE from: • Each record in one table can relate to many records in another table (1)	1
4(b)	THREE from: • Need to create a link table (1) • Is created using two one-to-many relationships (1) • Each primary key must exist as a foreign key (1st) • in the link table (1) (Only award once) • The primary key(s) link(s) to the foreign key(s) (1st) • in the link table (1) (only award once) • Not possible to create a single many-to-many relationship (1) • Only a theoretical type of relationship//cannot exist (1).	3

Mark scheme · 9626/12 Oct/Nov 2024 · Question 1

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Question	Answer	Marks
1(a)	THREE from: - Structure defines levels of access to records/data (1) - Uses <u>one-to-many</u> relationships (1) - Uses a structure (1) - Links are based on dependencies (1) - Links a number of records to one owner/parent (1) - The system is inflexible (1) - It is limited by using only one type of relationship (1st) so is confined to some very specific uses. (1)	3
1(b)	THREE from: - Records would be stored in a fixed/structured format (1st) that allows different levels of access (1) Up to TWO marks for examples (must be specific examples from the example) e.g. - The managing director can see/access all records/departments (1) - One director can ONLY see/access Department A and Department B (1) Up to THREE marks: - The managing director can only be a parent record (1) - A worker can be a child record (1) - A director/department can be both a parent and a child record (1)	3

Mark scheme · 9626/12 Oct/Nov 2024 · Question 8

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Question	Answer	Marks																		
8(a)	<table> <tr> <th>Field Name</th><th>Data Type</th><th>Example of Data</th></tr> <tr> <td>Number_in_Stock</td><td>Integer (1)</td><td>199</td></tr> <tr> <td>Location</td><td>String/alphanumeric/text (1)</td><td>07.44.03</td></tr> <tr> <td>Reorder_Level</td><td>Integer (1)</td><td>250</td></tr> <tr> <td>Price</td><td>Currency (1)</td><td>\$203.50</td></tr> <tr> <td>Supplier_Telephone_No.</td><td>String/alphanumeric/text(1)</td><td>07700900819</td></tr> </table>	Field Name	Data Type	Example of Data	Number_in_Stock	Integer (1)	199	Location	String/alphanumeric/text (1)	07.44.03	Reorder_Level	Integer (1)	250	Price	Currency (1)	\$203.50	Supplier_Telephone_No.	String/alphanumeric/text(1)	07700900819	5
Field Name	Data Type	Example of Data																		
Number_in_Stock	Integer (1)	199																		
Location	String/alphanumeric/text (1)	07.44.03																		
Reorder_Level	Integer (1)	250																		
Price	Currency (1)	\$203.50																		
Supplier_Telephone_No.	String/alphanumeric/text(1)	07700900819																		

Question	Answer	Marks
8(b)	SIX from: Three matched pairs: • Number_in_Stock – Type check (1st) must be numeric (1) • Number_in_Stock – Range check (1st) between 0 and 250 (1) • Price – Limit check (1st) must be more than zero (1) • Price – Type check (1st) must be numeric/currency (1) • Location – Format check (1st) Must be three pairs of digits separated by a full stop/NN.NN.NN (1) • Location – Length check (1st) must be 8 characters (1) • Reorder_Level – Type check (1st) must be numeric (1) • Supplier_Telephone_No – Length check (1st) must be 11 characters/digits (1).	6
8(c)	FIVE from: • Any awareness of normalising data (1) • Import the (two) tables into the relational database structure (1st) as separate tables (1) • Create/use a linking table (1) • Create/use/it needs Primary Keys for both/all tables (1) • Create/use/it needs Foreign Keys. (1st) These are needed to act as the link field between the tables (only award if BOTH of the previous two answers has been given) (1) • In the linking table (1) • Link the Primary Key(s) to the Foreign Key(s) (1) • Any awareness of one-to-many relationships (1)	5

Mark scheme · 9626/13 Oct/Nov 2024 · Question 2

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Question	Answer	Marks
2	SIX from, e.g.: • MIS is a computer-based system that provides managers with the tools to organise/evaluate/efficiently manage departments within an organisation (1) • It provides past, present and prediction information (answer must include TWO from) (1) • It is software based (1) • It includes a database/databases (1) • It includes a number of decision support systems (1) • It can be used to produce reports (1) • Can produce graphs/charts (1)	6

Question	Answer	Marks
3(a)	TWO from: • A database is in 1NF (accept list of 1NF principles) (1) • All the non-key attributes depend entirely on the primary key/it has no partial dependencies. (1)	2
3(b)	EIGHT from, e.g.: Benefits MAX 6 • Atomised data is easier to search on specific criteria (1) • Any change to one record which is needed can instantly be made to any related records//data editing is straightforward (1) • The database does not have redundant data (1) • File size is small/less storage needed (1) • There is no data duplication (1st) – so there are fewer errors in the data (1) – so there is less chance of storing incorrect copies of the data (1) Drawbacks MAX 6 • It can take longer to set up the database (1) • It is a more complex database (1st) • due to the relationships between tables (1) • Making data atomic may not always be the best solution (1ST) • such as (E.G.) date of birth can be separated into day, month and year but this may serve no purpose • Has more/multiple tables (1) • therefore setting up queries/reports/input forms can become more difficult (1) • making it difficult to keep track of data (1) • May require greater expertise (which may need to be bought in) (1).	8

Mark scheme · 9626/11 May/June 2025 · Question 10

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Question	Answer	Marks
10(a)	One mark per bullet point to a maximum of two marks. - Are used to summarise the contents of a table//summarise field values (1) - Also called Group-By queries/aggregate queries (1) - use aggregate functions (1) - They are used to extract aggregate of data items for a group of records - Uses functions such as//calculates (TWO of) SUM, AV(ERA)G(E), MIN, MAX, COUNT	2
10(b)	One mark per bullet point to a maximum of two marks. - Searches using more than one parameter value (1) - Searches on more than one field (1) - It often uses Boolean operands (1) - It is made up of (TWO of) AND, OR or NOT operands or a mixture of these (1)	2
10(c)	One mark per bullet point to a maximum of two marks. - A nested query is a query within another query (1) - It can be often referred to as a subquery (1) - Use the result of one query (1st) - as an input parameter of another (1) - Innermost subquery is executed first, then next level, until the main query is reached. (1)	2

Mark scheme · 9626/12 May/June 2025 · Question 7

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Question	Answer	Marks
7	One mark per bullet point to a maximum of five marks. - Indexed sequential access is a mixture of sequential access and direct/random access (1) - A set of hash tables known as indexes contain pointers into the records/index value is generated (1) - The index is a pointer to whereabouts on the disk the record is stored (1) - Individual records can be retrieved without having to search the entire file (1) - Data is organised into records which consist of fixed length fields (1) - Records can be found randomly using an index (1) - Indexes can be searched quickly, thereby allowing the database to access only the records it needs (1) - Searches use an index which will narrow down the records/data to be searched//focuses on a subset of records (1st) - Then that section of the file is searched sequentially to find the record required. (1)	5

Mark scheme · 9626/12 May/June 2025 · Question 9

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Question	Answer	Marks
9	Max 6 from three matched pairs • Different software on Eric's computer than his friends' computers (1st) – E.g. use generic file type (1) • Saved in a different/wrong (file) format (1st) – E.g. use generic file format (1) • Anti-Virus does not allow file to open/File has a virus (1st) – E.g. Eric needs to virus check the file before sending it (1) • File is password protected/encrypted and Eric didn't share the password/decryption key (1st) – E.g. Eric needs to share the password (1). • Wrong access rights/permissions assigned to the file (1st) – E.g. grant access rights/permissions (1)	6

Mark scheme · 9626/13 May/June 2025 · Question 12

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Question	Answer	Marks
12(a)	• A composite key is a primary key that uses the contents of two or more fields from a table to create a unique value. (1)	1
12(b)	• A compound key is a primary key that combines more than one foreign key to create a unique value. (1)	1

Mark scheme · 9626/11 and 9626/13 Oct/Nov 2025 · Question 7

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Question	Answer	Marks
7(a)	Two from: • A query that prompts the user for an input (1) • Allows the parameter/criteria to be changed each time (1) • Allows to search based on specific parameters/criteria the user needs at any given time (1)	2
7(b)	Three from: • Efficient way to search a database (1) • Reduces the number of queries saved//number of queries through which to search (1) • Reduced file size (1) • Requires little technical knowledge by end user (1) • Allows the end user flexibility/control (1) • Saving time in not creating a new query every time (1) • It provides a user-friendly way to interact with the database (1)	3

Question	Answer	Marks
15	Four from: • It is used in database design (1) • Describes the relationship between two tables (1) • It ensures that all data in a database remains consistent (1) • It ensures that it is all up to date (1) • It ensures that all foreign keys are dependent on the matching primary key (1) • Gives user a warning if they try to delete parent data (1) • Helps to reduce redundant data (1)	4

Question	Answer	Marks
7(a)	Four from: • Decide on what you want to learn/know (1) • Identify/prepare questions (1) • Decide on interviewer (1) • Identify population/decide which group of staff to interview (1) • Choose suitable date/time (1) • Choose suitable location (any indication of choice OR criteria) (1) • Select/invite attendees/employees (1) • Choose suitable timetable/length (1) • Prepare suitable recording methodology (1)	4
7(b)	Four from: Analysis • You can create complex reports (1st) – allows to combine data from multiple tables (1) • You can create complex queries (1st) – ... (1) • You can create dynamic queries (1st) – ... (1) Organisation • Quicker to search (1st) – Because data is held in tables (1) Consistency • Data is not repeated/reduced redundant data/only entered once (1st) – Quicker/easier to modify the data (1) – File size can be smaller/frees up (NOT “creates”) storage space (1) – Reduced chance for errors (1)	4

Question	Answer	Marks								
10(a)	THREE marks available: <table><tr><th>Data title</th><th>Most appropriate data type</th></tr><tr><td>employee_name</td><td>Text/alphanumeric (1)</td></tr><tr><td>hourly_rate_of_pay</td><td>Currency/numeric (1)</td></tr><tr><td>was_employee_born_in_france?</td><td>Boolean (1)</td></tr></table>	Data title	Most appropriate data type	employee_name	Text/alphanumeric (1)	hourly_rate_of_pay	Currency/numeric (1)	was_employee_born_in_france?	Boolean (1)	3
Data title	Most appropriate data type									
employee_name	Text/alphanumeric (1)									
hourly_rate_of_pay	Currency/numeric (1)									
was_employee_born_in_france?	Boolean (1)									
10(b)(i)	TWO marks available: First mark for identifying a suitable validation rule/test (1st) Second mark for a description of that test (1) • Range (1st) – to check value is BETWEEN two figures (1) • Type – to check data is in numeric/currency • Format check – to check entered as (e.g.) \$3.14//description of • Lookup check – to check against a list of values • Limit check – to check that is not below//above a certain figure • Character check – to check that is a decimal/real	2								

Question	Answer	Marks
10(b)(ii)	TWO marks available: First mark for identifying a suitable validation test (1st) Second mark for a description of that test (1) • Range – to check value is BETWEEN two figures • Type – to check data is in numeric/date • Format check – to check entered as (e.g.) dd/mm/yyyy • Limit check – to check that is not below/above a certain figure • Character check – to check that is an integer	2
10(c)	THREE marks available: • Age is based on the calculation between date of birth and today (1) • Age constantly increases with time//age changes every year (1) • A calculated field allows for a changing/dynamic figure (1) • The result shown//age is constantly accurate (1st) Reduced chance of human error resulting in the wrong number being stored (1)	3

Question	Answer	Marks	Guidance
5(a)	FOUR from: STRUCTURE - In a flat-file database, there will be one table// in a relational database there will be more than one table (1) ACCURACY (accept converse for each) - A relational database is less likely to have data entry errors (1) (For the same data,) a relational database will be smaller//save storage space (1) THE DATA - A relational database will have less duplicated data/redundant data (1) OR - a flat-file database can have the same data repeated (many times)//have lots of redundancy (1) - Each record in a relational database only has to be entered once//in a flat-file database, the same data must be entered every time it is needed (1) - Any mention of data integrity being applied to a relational database (1) TIME (accept converse for each) - Relational databases take less time to maintain/edit the data (1) - Relational database takes longer to edit the structure (1) - Querying a relational database is faster than querying a flat-file databases (1) COMPLEXITY (do not accept converse) - Relational database is more complex (1) CAN ALSO AWARD IF STATED SEPARATELY - Relational database can create complex queries (1) - Relational database can create complex reports (1)	4	Do Not accept - Less time/more time to plan/make - Easier to create/update - Any reference to files rather than tables

Question	Answer	Marks	Guidance
5(b)	Three marks available: Structure: - Identifies the validation type (1st) - Describes a rule (1) - States why it is important//why is it used//what error does it stop (1) e.g. - Presence check (1st) Each field has to contain an item of data//cannot be left empty (1) To check that data has been entered into the field//because the data is necessary (1)	3	

Question	Answer	Marks	Guidance
9(a)	ONE mark available: - A database that stores data in more than one linked table (1) - It has linked tables (1)	1	Do Not award - Entities - Multiple databases - Multiple locations - Answers such as ‘two tables with relationships’ Use of relationship here – mark as REP
9(b)(i)	TWO from: - Contains unique data (1) - Cannot contain a null value//must contain a value (1) - Can be based on one cell OR composite/based on two cells (1) (whole answer required) Only one per table (1)	2	
9(b)(ii)	ONE from: - It enables links to be formed between tables (1) - It uniquely identifies a record (1) - It may be used in a different table as a foreign key (1) - It maintains accuracy/consistency of data (1) - It allows for quick access to individual records (1)	1	

Question	Answer	Marks	Guidance
9(c)	ONE mark available: - A compound key is two or more pieces of data//a primary key (for an individual record) is a single piece of data (1) - A compound key is based on more than one column//a primary key is based on one column (1) - A compound key gives uniqueness from two columns of data (or more)//a primary key gives uniqueness from one column of data (1)	1	
9(d)	FOUR from: - All data must be atomic (1) - There must be a primary key (1) - There must be no duplicated rows/columns (1) - There must be only one value for each row of each column (1)	4	
9(e)	FOUR from: - Any change to one record which is needed can instantly be made to any related records/improved data consistency (1) - Provides referential integrity (1) - The database does not have redundant data (1) - Makes the file size small/less storage needed (1) - There is no/less data duplication/repeating data (1st) so there are fewer errors in the data (1) so there is less chance of storing incorrect copies of the data (1) - There is less data to modify (1) - Change (to data) can be made more quickly (1)	4	Do Not Award No repeat fields